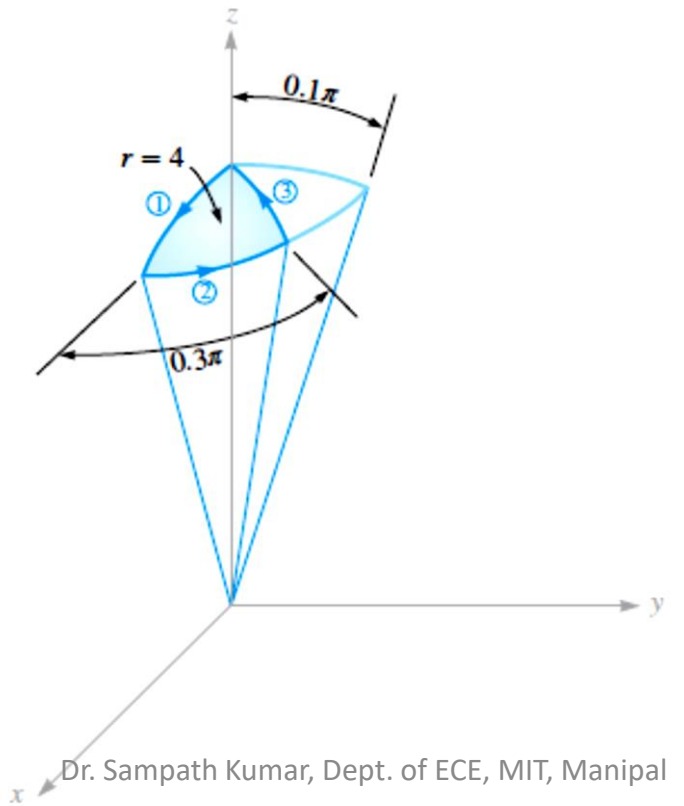
Magnetic Flux density

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Try this solved example from the textbook

EXAMPLE 7.3

A numerical example may help to illustrate the geometry involved in Stokes' theorem. Consider the portion of a sphere shown in Figure 7.17. The surface is specified by $r = 4, 0 \le \theta \le 0.1\pi, 0 \le \phi \le 0.3\pi$, and the closed path forming its perimeter is composed of three circular arcs. We are given the field $\mathbf{H} = 6r \sin\phi \mathbf{a}_r + 18r \sin\theta \cos\phi \mathbf{a}_\phi$ and are asked to evaluate each side of Stokes' theorem.



Curl Formula in Spherical

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

$$d\mathbf{S} = r^2 \sin\theta \, d\theta \, d\phi \, \mathbf{a}_r$$

22.2 A

$$\int H_\phi r \sin\theta \, d\phi$$

D7.6. Evaluate both sides of Stokes' theorem for the field $\mathbf{H} = 6xy\mathbf{a}_x - 3y^2\mathbf{a}_y$ A/m and the rectangular path around the region, $2 \le x \le 5, -1 \le y \le 1, z = 0$. Let the positive direction of $d\mathbf{S}$ be $\mathbf{a}_z$.

Ans. −126 A; −126 A

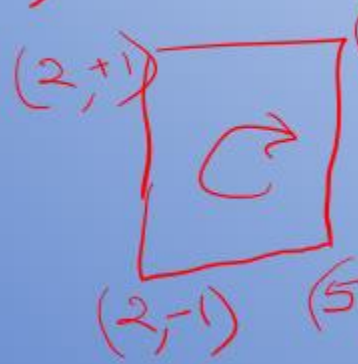
LHS $\int \nabla \times H = J \Rightarrow$

$$\begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy & -3y^2 & 0 \end{vmatrix} = \hat{a}_x(-6y) - \hat{a}_y \cdot 0 + \hat{a}_z(0 - 6x)$$

$$= (-6y\,\hat{a}_x - 6x\,\hat{a}_z)$$

$$I = \int_S J \cdot dS = \int_{y=-1}^{1} \int_{x=2}^{5} (-6y\hat{a}_x - 6x\,\hat{a}_z) \cdot dx\,dy\,\hat{a}_z = -126\ A$$

RHS $= \int H \cdot dl$



$$\Rightarrow \int_{2}^{5} H_x|_{y=1}\,dx + \int_{-1}^{1} H_y|_{x=5}\,dy + \int_{5}^{2} H_x|_{y=-1}\,dx + \int_{1}^{-1} H_y|_{x=2}\,dy$$

$$= -(63 + 63) = -126\ A$$

In free space, we define the *magnetic flux density* **B** as

$$\mathbf{B} = \mu_0 \mathbf{H} \quad \text{(free space only)}$$

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S} \text{ Wb}$$

webers per square meter (Wb/m^2) or tesla (T).

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

The name given to μ_0 is the *permeability* of free space.

$$\mu = \mu_0 \mu_r$$

$$\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q$$

The magnetic flux lines are closed and do not terminate on a “magnetic charge.“

For this reason, Gauss’s law for the magnetic field is

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

and application of the divergence theorem shows us that

$$\nabla \cdot \mathbf{B} = 0$$

Maxwell’s four equations as they apply to static electric fields and steady magnetic fields.

$$\nabla \cdot \mathbf{D} = \rho_v$$
$$\nabla \times \mathbf{E} = 0$$
$$\nabla \times \mathbf{H} = \mathbf{J}$$
$$\nabla \cdot \mathbf{B} = 0$$

Incomplete

Maxwell's four equations as they apply to static electric fields and steady magnetic fields

	Point form	Integral form
1	$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dv$
2	$\nabla \times \mathbf{E} = 0$	$\oint \mathbf{E} \cdot d\mathbf{L} = 0$
3	$\nabla \times \mathbf{H} = \mathbf{J}$	$\oint \mathbf{H} \cdot d\mathbf{L} = I = \int_S \mathbf{J} \cdot d\mathbf{S}$
4	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{S} = 0$

Note: For time varying fields, 2 and 3 are different


Thank
you